Chapter 4

Convex sets and convex functions.

Abstract:
Convexity is the structural assumption that turns many optimization problems from "hard in principle" into "tractable in practice". In this chapter we study: convex *sets* and convex *functions*. We focus on geometric intuition, standard examples, and closure properties (operations that preserve convexity) that let us build new convex objects from old ones.

4.1 Introduction

Convex sets and functions appear throughout optimization: constrained problems are built from convex feasible regions, and many machine-learning losses are convex (or nearly so) in the parameters. Later, when we introduce constrained optimization and projections, we will repeatedly rely on the basic facts collected here.

4.2 Convex sets

4.2.1 Definition and geometric picture

Definition 4.1 (Convex set)**.** Let U be a real vector space and let $\mathcal{Q} \subseteq U$. We say that $\mathcal{Q}$ is *convex* if for all $\mathbf{x}, \mathbf{y} \in \mathcal{Q}$ and all $\lambda \in [0, 1]$,

$$\lambda \mathbf{x} + (1 - \lambda)\mathbf{y} \in \mathcal{Q}.$$

Geometrically, convexity means that for any two points in the set, the whole line segment between them stays in the set (see fig. 4.1).

4.2.2 Standard examples

Many convex sets can be checked directly from the definition by "plugging in" $\lambda\mathbf{x}+(1-\lambda)\mathbf{y}$.

Example 4.1 (Simple convex sets)**.**

- The empty set $\varnothing$ and any singleton $\{\mathbf{a}\}$ are convex.
- $\{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{a_i}^\top \mathbf{x} \ \ [\leq | = | \geq | < | >] \ \ b_i, i \in I\}$, where I is uncountable, finite or

convex nonconvex

Figure 4.1: A set is convex if it contains the entire segment between any two of its point.

countable set. Special cases:

- Any polyhedron (finite intersection of halfspaces/hyperplanes) is convex:

$$\{\mathbf{x} \in \mathbb{R}^n : \mathbf{A}\mathbf{x} \leq \mathbf{b},\ \mathbf{C}\mathbf{x} = \mathbf{d}\}.$$

- The probability simplex is convex:

$$\Delta_n := \Big\{\mathbf{x} \in \mathbb{R}^n :\ \mathbf{x} \geq 0,\ \sum_{i=1}^n x_i = 1\Big\}.$$

Example 4.2 (Balls)**.** For a norm $\|\cdot\|$ on $\mathbb{R}^n$, the ball

$$\mathcal{B}_r(\mathbf{a}) := \{\mathbf{x} \in \mathbb{R}^n :\ \|\mathbf{x} - \mathbf{a}\| \leq r\}$$

is convex: if $\|\mathbf{x} - \mathbf{a}\| \leq r$ and $\|\mathbf{y} - \mathbf{a}\| \leq r$, then by the triangle inequality and homogeneity,

$$\|\lambda(\mathbf{x} - \mathbf{a}) + (1 - \lambda)(\mathbf{y} - \mathbf{a})\| \leq \lambda\|\mathbf{x} - \mathbf{a}\| + (1 - \lambda)\|\mathbf{y} - \mathbf{a}\| \leq r.$$

Example 4.3 (Positive (semi)definite matrices are convex)**.** Let $\mathbb{S}^n := \{\mathbf{A} \in \mathbb{R}^{n\times n} : \mathbf{A} = \mathbf{A}^\top\}$, and define the cones

$$\mathbb{S}^n_{++} := \{\mathbf{A} \in \mathbb{S}^n : \mathbf{A} \succ 0\}, \qquad \mathbb{S}^n_{+} := \{\mathbf{A} \in \mathbb{S}^n : \mathbf{A} \succeq 0\}.$$

Both sets are convex. For example, if $\mathbf{A}, \mathbf{B} \in \mathbb{S}^n_{++}$ and $\lambda \in [0, 1]$, then for any nonzero $\mathbf{x}$,

$$\mathbf{x}^\top(\lambda\mathbf{A} + (1 - \lambda)\mathbf{B})\mathbf{x} = \lambda\,\mathbf{x}^\top\mathbf{A}\mathbf{x} + (1 - \lambda)\,\mathbf{x}^\top\mathbf{B}\mathbf{x} > 0,$$

so $\lambda\mathbf{A} + (1-\lambda)\mathbf{B} \in \mathbb{S}^n_{++}$. The same argument with ≥ 0 shows convexity of $\mathbb{S}^n_{+}$.

4.2.3 Operations that preserve convexity of sets

Convex sets are closed under several basic constructions. These "closure rules" are practical: they let us recognize convex feasible regions without going back to the definition every time.

Claim 4.1 (Intersection)**.** If $\{\mathcal{Q}_i\}_{i\in I}$ is a family of convex sets, then $\bigcap_{i\in I}\mathcal{Q}_i$ is convex.

Proof: Take $\mathbf{x}, \mathbf{y} \in \bigcap_{i\in I}\mathcal{Q}_i$. Then $\mathbf{x}, \mathbf{y} \in \mathcal{Q}_i$ for all i, hence $\lambda\mathbf{x} + (1-\lambda)\mathbf{y} \in \mathcal{Q}_i$ for all i by convexity of each $\mathcal{Q}_i$. Therefore $\lambda\mathbf{x} + (1-\lambda)\mathbf{y} \in \bigcap_{i\in I}\mathcal{Q}_i$. □

Claim 4.2 (Affine images and preimages)**.** Let $A : U \to V$ be an affine map, $A(\mathbf{x}) = \mathbf{L}\mathbf{x} + \mathbf{b}$.

- If $\mathcal{Q} \subseteq U$ is convex, then its image $A(\mathcal{Q}) := \{A(\mathbf{x}) : \mathbf{x} \in \mathcal{Q}\} \subseteq V$ is convex.
- If $\mathcal{S} \subseteq V$ is convex, then its preimage $A^{-1}(\mathcal{S}) := \{\mathbf{x} \in U : A(\mathbf{x}) \in \mathcal{S}\} \subseteq U$ is convex.

Proof: For the image: take $\mathbf{u} = A(\mathbf{x})$ and $\mathbf{v} = A(\mathbf{y})$ with $\mathbf{x}, \mathbf{y} \in \mathcal{Q}$. Then

$$\lambda\mathbf{u} + (1-\lambda)\mathbf{v} = \lambda(\mathbf{L}\mathbf{x} + \mathbf{b}) + (1-\lambda)(\mathbf{L}\mathbf{y} + \mathbf{b}) = \mathbf{L}(\lambda\mathbf{x} + (1-\lambda)\mathbf{y}) + \mathbf{b}.$$

Since $\lambda\mathbf{x} + (1-\lambda)\mathbf{y} \in \mathcal{Q}$, the right-hand side lies in $A(\mathcal{Q})$.

For the preimage: if $A(\mathbf{x}), A(\mathbf{y}) \in \mathcal{S}$, then by convexity of $\mathcal{S}$ and affinity of A,

$$A(\lambda\mathbf{x} + (1-\lambda)\mathbf{y}) = \lambda A(\mathbf{x}) + (1-\lambda)A(\mathbf{y}) \in \mathcal{S}.$$

Hence $\lambda\mathbf{x} + (1-\lambda)\mathbf{y} \in A^{-1}(\mathcal{S})$. □

Claim 4.3 (Cartesian products)**.** If $\mathcal{Q}_1 \subseteq U_1$ and $\mathcal{Q}_2 \subseteq U_2$ are convex, then $\mathcal{Q}_1 \times \mathcal{Q}_2$ is convex in $U_1 \times U_2$.

Proof: If $(\mathbf{x}_1, \mathbf{x}_2), (\mathbf{y}_1, \mathbf{y}_2) \in \mathcal{Q}_1 \times \mathcal{Q}_2$, then

$$\lambda(\mathbf{x}_1, \mathbf{x}_2) + (1-\lambda)(\mathbf{y}_1, \mathbf{y}_2) = (\lambda\mathbf{x}_1 + (1-\lambda)\mathbf{y}_1,\ \lambda\mathbf{x}_2 + (1-\lambda)\mathbf{y}_2),$$

and each component lies in the corresponding set by convexity. □

Example 4.4 (Intersections, products, and affine preimages)**.**

- The set
$$\left\{\mathbf{x} \in \mathbb{R}^2 :\ x_1^2 + x_2^2 \leq 1,\ x_2 \geq 0\right\}$$
is convex because it is an intersection of two convex sets: the Euclidean unit ball and a halfspace.
- The set
$$\left\{\mathbf{x} \in \mathbb{R}^3 :\ x_1^2 + x_2^2 \leq 1,\ |x_3| \leq 1\right\}$$

is convex because it is a Cartesian product:

$$\Big\{(x_1, x_2) \in \mathbb{R}^2 : \; x_1^2 + x_2^2 \le 1\Big\} \times \Big\{x_3 \in \mathbb{R} : \; |x_3| \le 1\Big\}.$$

- Fix $\mathbf{A}_1, \ldots, \mathbf{A}_k, \mathbf{B} \in \mathbb{S}^n$ and consider

$$\mathcal{Q} := \Big\{\mathbf{x} \in \mathbb{R}^k : \; \mathbf{B} - \sum_{i=1}^{k} x_i \mathbf{A}_i \succeq 0\Big\}.$$

  The map $\mathbf{x} \mapsto \mathbf{B} - \sum_{i=1}^{k} x_i \mathbf{A}_i$ is affine, and image $\in \mathbb{S}_+^n$ is convex; therefore $\mathcal{Q}$ is convex due to Claim 4.2

4.3 Convex hulls and cones

4.3.1 Convex combinations and convex hull

Definition 4.2 (Convex combination)**.** Given points $\mathbf{x}_1, \ldots, \mathbf{x}_m$ in a vector space and weights $\lambda_1, \ldots, \lambda_m \ge 0$ with $\sum_{i=1}^m \lambda_i = 1$, the point

$$\sum_{i=1}^{m} \lambda_i \mathbf{x}_i$$

is called a *convex combination* of $\{\mathbf{x}_i\}_{i=1}^m$.

Claim 4.4 (A set is convex iff it contains all finite convex combinations)**.** Let $\mathcal{M} \subseteq U$. Then $\mathcal{M}$ is convex if and only if for every $m \ge 2$, every $\mathbf{x}_1, \ldots, \mathbf{x}_m \in \mathcal{M}$, and every $\lambda_i \ge 0$ with $\sum_{i=1}^m \lambda_i = 1$, we have $\sum_{i=1}^m \lambda_i \mathbf{x}_i \in \mathcal{M}$.

Proof: If $\mathcal{M}$ contains all finite convex combinations, then the case $m = 2$ is exactly the definition of convexity.

Conversely, assume $\mathcal{M}$ is convex in the sense of $m = 2$. We prove the statement for general m by induction. The case $m = 2$ holds by assumption. Suppose it holds for m, and consider $m+1$ points with weights $\lambda_1, \ldots, \lambda_{m+1} \ge 0$, $\sum_{i=1}^{m+1} \lambda_i = 1$. If $\lambda_{m+1} = 1$ the claim is trivial; otherwise define $\alpha := \lambda_{m+1} \in [0, 1)$ and

$$\mathbf{z} := \sum_{i=1}^{m} \frac{\lambda_i}{1-\alpha} \mathbf{x}_i.$$

By the induction hypothesis, $\mathbf{z} \in \mathcal{M}$. Then

$$\sum_{i=1}^{m+1} \lambda_i \mathbf{x}_i = (1-\alpha)\mathbf{z} + \alpha \mathbf{x}_{m+1} \in \mathcal{M}$$

by convexity of $\mathcal{M}$. $\square$

Definition 4.3 (Convex hull (see fig. 4.2))**.** For $\mathcal{M} \subseteq U$, the *convex hull* of $\mathcal{M}$, denoted $\operatorname{conv}(\mathcal{M})$, is the smallest convex set that contains $\mathcal{M}$. Equivalently,

$$\operatorname{conv}(\mathcal{M}) = \Big\{ \sum_{i=1}^{m} \lambda_i \mathbf{x}_i : \ m \in \mathbb{N}, \ \mathbf{x}_i \in \mathcal{M}, \ \lambda_i \geq 0, \ \sum_{i=1}^{m} \lambda_i = 1 \Big\}.$$

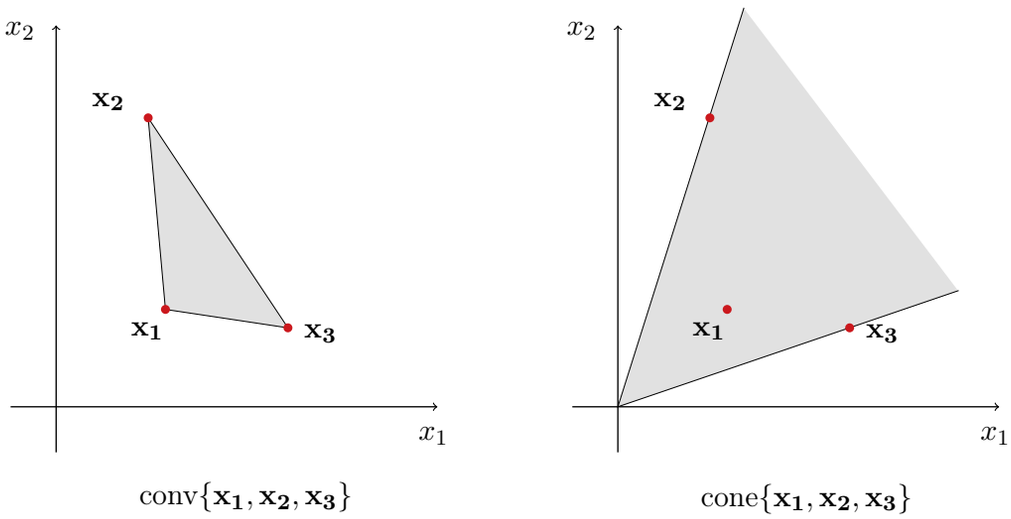

Figure 4.2: Convex hull versus conic hull

4.3.2 Cones and convex cones

Definition 4.4 (Cone and convex cone)**.** A set $\mathcal{K} \subseteq U$ is a *cone* if

$$\mathbf{x} \in \mathcal{K}, \ \alpha \geq 0 \quad \Longrightarrow \quad \alpha \mathbf{x} \in \mathcal{K}.$$

It is a *convex cone* if, in addition,

$$\mathbf{x}, \mathbf{y} \in \mathcal{K}, \ \alpha, \beta \geq 0 \quad \Longrightarrow \quad \alpha \mathbf{x} + \beta \mathbf{y} \in \mathcal{K}.$$

Definition 4.5 (Conic hull (see fig. 4.2))**.** For $\mathcal{M} \subseteq U$, the (conic) hull of $\mathcal{M}$ is

$$\operatorname{cone}(\mathcal{M}) := \Big\{ \sum_{i=1}^{m} \lambda_i \mathbf{x}_i : \ m \in \mathbb{N}, \ \mathbf{x}_i \in \mathcal{M}, \ \lambda_i \geq 0 \Big\}.$$

Example 4.5 (Second-order cone (see fig. 4.3))**.** The set

$$\mathcal{K}_{\text{soc}} := \{ (\mathbf{x}, t) \in \mathbb{R}^n \times \mathbb{R} : \ \|\mathbf{x}\|_2 \leq t \}$$

is a convex cone. Indeed, for $\alpha, \beta \geq 0$,

$$\|\alpha\mathbf{x} + \beta\mathbf{y}\|_2 \leq \alpha\|\mathbf{x}\|_2 + \beta\|\mathbf{y}\|_2 \leq \alpha t_x + \beta t_y,$$

so $(\alpha\mathbf{x} + \beta\mathbf{y}, \alpha t_x + \beta t_y) \in \mathcal{K}_{\text{soc}}$ whenever $(\mathbf{x}, t_x), (\mathbf{y}, t_y) \in \mathcal{K}_{\text{soc}}$.

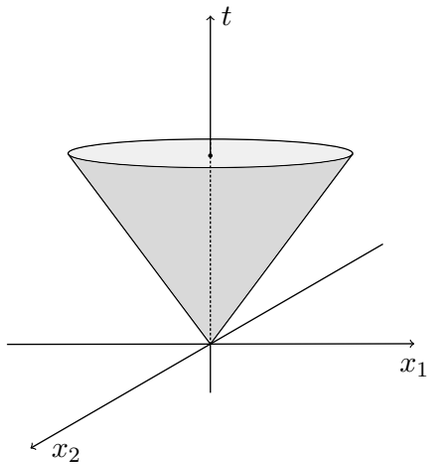

Figure 4.3: A 2D visualization of the second-order cone

Example 4.6. Consider the set

$$\mathcal{Q} := \{\mathbf{x} \in \mathbb{R}^n : \ \|\mathbf{A}\mathbf{x} + \mathbf{b}\|_2 \leq \mathbf{c}^\top\mathbf{x} + d\}, \qquad \mathbf{A} \in \mathbb{R}^{m\times n},\ \mathbf{b} \in \mathbb{R}^m,\ \mathbf{c} \in \mathbb{R}^n,\ d \in \mathbb{R}.$$

Introduce the augmented variable $\tilde{\mathbf{x}} := (\mathbf{x}, 1) \in \mathbb{R}^{n+1}$ and matrices

$$\tilde{\mathbf{A}} := [\mathbf{A}\ \mathbf{b}] \in \mathbb{R}^{m\times(n+1)}, \qquad \tilde{\mathbf{c}} := (\mathbf{c}, d) \in \mathbb{R}^{n+1}.$$

Then $\|\mathbf{A}\mathbf{x} + \mathbf{b}\|_2 \leq \mathbf{c}^\top\mathbf{x} + d$ becomes

$$\|\tilde{\mathbf{A}}\tilde{\mathbf{x}}\|_2 \leq \tilde{\mathbf{c}}^\top\tilde{\mathbf{x}}.$$

Define the set

$$\mathcal{M} := \{\tilde{\mathbf{x}} \in \mathbb{R}^{n+1} : \ \|\tilde{\mathbf{A}}\tilde{\mathbf{x}}\|_2 \leq \tilde{\mathbf{c}}^\top\tilde{\mathbf{x}}\}.$$

By the same triangle-inequality argument as in the second-order cone example, $\mathcal{M}$ is a convex cone. Finally, $\mathcal{Q}$ is the affine preimage (a "slice") of $\mathcal{M}$:

$$\mathcal{Q} = \{\mathbf{x} \in \mathbb{R}^n : \ (\mathbf{x}, 1) \in \mathcal{M}\},$$

so $\mathcal{Q}$ is convex by the affine-preimage rule from Section 4.2.3.

4.4 Convex functions

4.4.1 Definition and basic examples

We already introduced convexity of functions in Section 1.5.1. We repeat the definition here.

Definition 4.6 (Convex function)**.** Let $\mathcal{Q} \subseteq \mathbb{R}^n$ be a convex set and let $f : \mathcal{Q} \to \mathbb{R}$. We say that f is *convex* if for all $\mathbf{x}, \mathbf{y} \in \mathcal{Q}$ and $\lambda \in [0, 1]$,

$$f(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}).$$

If the inequality is strict for $\mathbf{x} \neq \mathbf{y}$ and $\lambda \in (0, 1)$, then f is *strictly convex*.

Example 4.7 (Three important convex functions)**.**

- Any affine function $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x} + b$ is convex (and concave).
- Any norm $\|\mathbf{x}\|$ is convex.
- If $\mathcal{Q} \subseteq \mathbb{R}^n$ is convex, the *indicator function*

$$\iota_{\mathcal{Q}}(\mathbf{x}) := \begin{cases} 0, & \mathbf{x} \in \mathcal{Q}, \\ +\infty, & \mathbf{x} \notin \mathcal{Q}, \end{cases}$$

  is convex.

Claim 4.5 (Jensen's inequality)**.** Let $\mathcal{Q} \subseteq \mathbb{R}^n$ be convex and let $f : \mathcal{Q} \to \mathbb{R}$ be convex. Then for any $\mathbf{x}_1, \ldots, \mathbf{x}_m \in \mathcal{Q}$ and any weights $\lambda_i \geq 0$ with $\sum_{i=1}^m \lambda_i = 1$,

$$f\Big(\sum_{i=1}^m \lambda_i \mathbf{x}_i\Big) \leq \sum_{i=1}^m \lambda_i f(\mathbf{x}_i).$$

Proof: For $m = 2$ this is the definition of convexity. Assume the statement holds for m, and consider $m + 1$ points with weights $\lambda_1, \ldots, \lambda_{m+1} \geq 0$, $\sum_{i=1}^{m+1} \lambda_i = 1$. If $\lambda_{m+1} = 1$ the claim is trivial; otherwise define $\alpha := \lambda_{m+1} \in [0, 1)$ and

$$\mathbf{z} := \sum_{i=1}^m \frac{\lambda_i}{1 - \alpha} \mathbf{x}_i.$$

By the induction hypothesis, $f(\mathbf{z}) \leq \sum_{i=1}^m \frac{\lambda_i}{1-\alpha} f(\mathbf{x}_i)$. Using convexity of f once more,

$$f\Big(\sum_{i=1}^{m+1} \lambda_i \mathbf{x}_i\Big) = f\big((1 - \alpha)\mathbf{z} + \alpha \mathbf{x}_{m+1}\big) \leq (1 - \alpha) f(\mathbf{z}) + \alpha f(\mathbf{x}_{m+1}) \leq \sum_{i=1}^{m+1} \lambda_i f(\mathbf{x}_i),$$

which completes the induction. □

4.4.2 Epigraphs and sublevel sets

Definition 4.7 (Epigraph (see fig. 4.4))**.** For a function $f : \mathcal{Q} \to \mathbb{R}$, its *epigraph* is

$$\operatorname{epi} f := \{(\mathbf{x}, t) \in \mathcal{Q} \times \mathbb{R} : \ f(\mathbf{x}) \leq t\}.$$

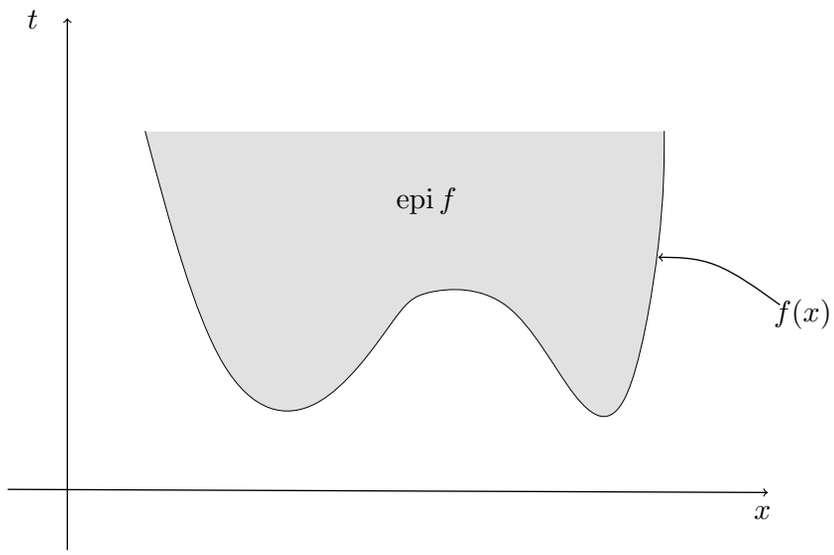

Figure 4.4: Epigraph of an arbitrary function.

Theorem 4.1 (Epigraph criterion)**.** A function f is convex if and only if $\operatorname{epi} f$ is a convex set.

Proof: Assume f is convex. Take $(\mathbf{x}, t_x), (\mathbf{y}, t_y) \in \operatorname{epi} f$, i.e., $f(\mathbf{x}) \leq t_x$ and $f(\mathbf{y}) \leq t_y$. By convexity of f,

$$f(\lambda\mathbf{x} + (1-\lambda)\mathbf{y}) \leq \lambda f(\mathbf{x}) + (1-\lambda) f(\mathbf{y}) \leq \lambda t_x + (1-\lambda) t_y,$$

so $(\lambda\mathbf{x} + (1-\lambda)\mathbf{y},\ \lambda t_x + (1-\lambda) t_y) \in \operatorname{epi} f$. Hence $\operatorname{epi} f$ is convex.

Conversely, assume $\operatorname{epi} f$ is convex. For any $\mathbf{x}, \mathbf{y} \in \mathcal{Q}$, the points $(\mathbf{x}, f(\mathbf{x}))$ and $(\mathbf{y}, f(\mathbf{y}))$ lie in $\operatorname{epi} f$. Convexity of the epigraph implies $(\lambda\mathbf{x} + (1-\lambda)\mathbf{y},\ \lambda f(\mathbf{x}) + (1-\lambda) f(\mathbf{y})) \in \operatorname{epi} f$, which is exactly the defining inequality of convexity of f. $\square$

Example 4.8 (Epigraph of a norm and the second-order cone)**.** For the Euclidean norm, the epigraph is exactly the second-order cone:

$$\operatorname{epi} \|\cdot\|_2 = \{(\mathbf{x}, t) \in \mathbb{R}^n \times \mathbb{R} : \ \|\mathbf{x}\|_2 \leq t\}.$$

Claim 4.6 (Sublevel sets of convex functions are convex)**.** If f is convex, then for every $\alpha \in \mathbb{R}$ the sublevel set

$$\{\mathbf{x} \in \mathcal{Q} : \ f(\mathbf{x}) \leq \alpha\}$$

is convex.

Proof: If $f(\mathbf{x}) \leq \alpha$ and $f(\mathbf{y}) \leq \alpha$, then by convexity, $f(\lambda\mathbf{x}+(1-\lambda)\mathbf{y}) \leq \lambda\alpha+(1-\lambda)\alpha = \alpha$. □

Remark (quasi-convexity). The converse statement is false in general: there exist nonconvex functions whose sublevel sets are all convex (e.g. x^3). Such functions are called *quasi-convex*.

4.4.3 Differential criteria

The next two characterizations connect convexity to gradients and Hessians. We stated these in Claim 1.3, but for consistency repeat here.

Claim 4.7 (First-order criterion)**.** Let $f \in C^1$ on a convex set $\mathcal{Q}$. Then f is convex if and only if for all $\mathbf{x}, \mathbf{y} \in \mathcal{Q}$,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}).$$

Claim 4.8 (Second-order criterion)**.** Let $f \in C^2$ on a convex set $\mathcal{Q}$. Then f is convex if and only if $\nabla^2 f(\mathbf{x}) \succeq 0$ for all $\mathbf{x} \in \mathcal{Q}$.

Example 4.9 (The exponential is convex)**.** For $f(x) = e^x$ on $\mathbb{R}$, we have $f''(x) = e^x > 0$ for all x, hence f is convex by Claim 4.8.

Example 4.10 (The log-sum-exp function is convex)**.** Define $f : \mathbb{R}^n \to \mathbb{R}$ by

$$f(\mathbf{x}) = \log\Big(\sum_{i=1}^n e^{x_i}\Big).$$

Let $s(\mathbf{x}) := \sum_{i=1}^n e^{x_i}$ and define the softmax weights

$$\pi_i(\mathbf{x}) := \frac{e^{x_i}}{s(\mathbf{x})}, \qquad \boldsymbol{\pi}(\mathbf{x}) \in \mathbb{R}^n, \ \sum_{i=1}^n \pi_i(\mathbf{x}) = 1.$$

Then $\nabla f(\mathbf{x}) = \boldsymbol{\pi}(\mathbf{x})$ and

$$\nabla^2 f(\mathbf{x}) = \operatorname{diag}(\boldsymbol{\pi}(\mathbf{x})) - \boldsymbol{\pi}(\mathbf{x})\boldsymbol{\pi}(\mathbf{x})^\top.$$

For any $\mathbf{d} \in \mathbb{R}^n$,

$$\mathbf{d}^\top \nabla^2 f(\mathbf{x})\, \mathbf{d} = \sum_{i=1}^n \pi_i(\mathbf{x})\, d_i^2 - \Big(\sum_{i=1}^n \pi_i(\mathbf{x})\, d_i\Big)^2 \geq 0,$$

where the inequality follows from Cauchy–Schwarz: $\left(\sum_i \pi_i d_i\right)^2 \leq \left(\sum_i \pi_i\right)\left(\sum_i \pi_i d_i^2\right) = \sum_i \pi_i d_i^2$. Hence $\nabla^2 f(\mathbf{x}) \succeq 0$ and f is convex by Claim 4.8.

4.4.4 Operations that preserve convexity of functions

Claim 4.9 (Convexity-preserving operations)**.** Let $\{f_i\}_{i\in I}$ be convex functions on a common convex domain, where I is an index set (finite, noncountable or countable), and let $c_i \geq 0$.

- (Nonnegative sums) $f(\mathbf{x}) = \sum_{i\in I} c_i f_i(\mathbf{x})$ is convex.
- (Affine composition) If f is convex and $A(\mathbf{x}) = \mathbf{L}\mathbf{x} + \mathbf{b}$ is affine, then $f \circ A$ is convex.
- (Pointwise supremum) $f(\mathbf{x}) = \sup_{i\in I} f_i(\mathbf{x})$ is convex.
- (Monotone composition) If f is convex and $g : \mathbb{R} \to \mathbb{R}$ is convex and nondecreasing, then $g \circ f$ is convex.
- (Partial minimization) If $f(\mathbf{x}, \mathbf{y})$ is convex jointly in $(\mathbf{x}, \mathbf{y})$, then

$$\varphi(\mathbf{x}) = \inf_{\mathbf{y}} f(\mathbf{x}, \mathbf{y})$$

  is convex.

Example 4.11 (Convex ML losses: multinomial logistic regression)**.** Let $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$ be data with classes $y_i \in \{1, \ldots, K\}$ and parameters $\mathbf{W} = [\mathbf{w}_1^\top; \ldots; \mathbf{w}_K^\top] \in \mathbb{R}^{K\times d}$. The negative log-likelihood (up to constants) can be written as

$$F(\mathbf{W}) = \sum_{i=1}^N \Big(-\mathbf{w}_{y_i}^\top \mathbf{x}_i + \log \sum_{j=1}^K \exp\Big(\mathbf{w}_j^\top \mathbf{x}_i\Big)\Big).$$

Each term $-\mathbf{w}_{y_i}^\top \mathbf{x}_i$ is affine, hence convex. The log-sum-exp term is convex by the previous example, and the map $\mathbf{W} \mapsto (\mathbf{w}_1^\top \mathbf{x}_i, \ldots, \mathbf{w}_K^\top \mathbf{x}_i)$ is affine. Therefore each summand is convex, and so is F by the nonnegative-sum rule.

Example 4.12 (The function $\|\mathbf{x}\|^p$ is convex for $p \geq 1$)**.** Let $f(\mathbf{x}) = \|\mathbf{x}\|$ (convex) and $g(t) = t^p$ on $t \geq 0$. For $p \geq 1$, g is convex and nondecreasing on $\mathbb{R}_+$. Hence $g \circ f(\mathbf{x}) = \|\mathbf{x}\|^p$ is convex.

Example 4.13 (Largest eigenvalue is convex)**.** For $\mathbf{A} \in \mathbb{S}^n$, the largest eigenvalue admits the variational representation

$$\lambda_{\max}(\mathbf{A}) = \max_{\|\mathbf{x}\|_2=1} \mathbf{x}^\top \mathbf{A}\mathbf{x}.$$

For fixed $\mathbf{x}$, the map $\mathbf{A} \mapsto \mathbf{x}^\top \mathbf{A}\mathbf{x}$ is linear in $\mathbf{A}$. Hence $\lambda_{\max}$ is a pointwise supremum

of linear functions of $\mathbf{A}$ and is therefore convex. Similarly,

$$\lambda_{\min}(\mathbf{A}) = \min_{\|\mathbf{x}\|_2=1} \mathbf{x}^\top \mathbf{A}\mathbf{x} = -\lambda_{\max}(-\mathbf{A})$$

is concave.

Example 4.14 (Distance to a convex set is convex)**.** Let $\mathcal{Q} \subseteq \mathbb{R}^n$ be convex and define the distance

$$\operatorname{dist}(\mathbf{x}, \mathcal{Q}) := \inf_{\mathbf{y}\in\mathcal{Q}} \|\mathbf{x} - \mathbf{y}\|.$$

Using the indicator function from Section 4.4.1,

$$\operatorname{dist}(\mathbf{x}, \mathcal{Q}) = \inf_{\mathbf{y}\in\mathbb{R}^n} \big(\|\mathbf{x} - \mathbf{y}\| + \iota_{\mathcal{Q}}(\mathbf{y})\big).$$

The function $(\mathbf{x}, \mathbf{y}) \mapsto \|\mathbf{x} - \mathbf{y}\|$ is convex, and $\iota_{\mathcal{Q}}$ is convex when $\mathcal{Q}$ is convex. Due to partial minimization operation the infimum over $\mathbf{y}$ preserves convexity, so $\mathbf{x} \mapsto \operatorname{dist}(\mathbf{x}, \mathcal{Q})$ is convex.

Example 4.15 (Input-convex neural networks (ICNNs))**.** In some applications we want to represent a convex function by a neural network, so that minimizing it is a convex optimization problem. For example, in control or reinforcement learning one may wish to choose a deterministic action by solving $\mathbf{a}^\star \in \arg\max_{\mathbf{a}} Q_\theta(\mathbf{s}, \mathbf{a})$ for a fixed state $\mathbf{s}$. If we model $Q_\theta(\mathbf{s}, \mathbf{a}) = -f_\theta(\mathbf{s}, \mathbf{a})$ with $f_\theta(\mathbf{s}, \cdot)$ convex, then greedy action selection reduces to minimizing $f_\theta(\mathbf{s}, \mathbf{a})$ and gives deterministic answer. Therefore, we don't stuck in the local minimum which is a crucial thing in this case.

One simple way to ensure convexity in $\mathbf{a}$ is an *input-convex neural network* (ICNN). Consider a feedforward network with hidden variables $\mathbf{z}_0 = \mathbf{0}$ and

$$\mathbf{z}_{i+1} = g_i\big(\mathbf{W}_i^z \mathbf{z}_i + \mathbf{W}_i^a \mathbf{a} + \mathbf{b}_i\big), \qquad i = 0, \dots, N-1,$$

where each g_i is applied componentwise. Assume that

- each activation $g_i : \mathbb{R} \to \mathbb{R}$ is convex and nondecreasing, and
- every entry of $\mathbf{W}_i^z$ is nonnegative: $(\mathbf{W}_i^z)_{pq} \geq 0$.

Then every component of $\mathbf{z}_i(\mathbf{a})$ is convex in $\mathbf{a}$, hence the scalar output $f_\theta(\mathbf{s}, \mathbf{a}) = \mathbf{z}_N(\mathbf{a})$ (take the last layer to have size 1) is convex in $\mathbf{a}$. The proof is an induction: nonnegative sums and affine additions preserve convexity, and composing with a convex nondecreasing map preserves convexity by the monotone-composition rule in Section 4.4.4.

In practice one often enforces $\mathbf{W}_i^z \geq 0$ by a reparameterization, for example

$$(\mathbf{W}_i^z)_{pq} = \log\Big(1 + \exp\Big((\tilde{\mathbf{W}}_i^z)_{pq}\Big)\Big),$$

so that the entries are automatically nonnegative.